A STUDY OF INFINTE PRODUCT EXPANSION FOR $\sin(\pi z)$

Graduate Complex Analysis

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PART 1: $\sin(\pi z)$

First we shall consider constructing some function *G(z)* that contains roots, *r*, at every integer along the negative and positive real axis of the complex plane, $r \in \mathbb{Z}$.

$$z = 0, \quad z = 1, \quad z = 2, \quad z = 3..., \quad z = n$$

If we take each equation above and divide by the root, with the exception of $z = 0$, each equation will have unity on the right side.

$$z = 1, \quad \frac{z}{2} = 1, \quad \frac{z}{3} = 1, ... \quad \frac{z}{n} = 1$$

If we bring all terms to one side we obtain,

$$1 - z = 0, \quad 1 - \frac{z}{2} = 0, \quad 1 - \frac{z}{3} = 0, ... \quad 1 - \frac{z}{n} = 0$$

The product of all the above equations yields,

$$(1 - z)\left(1 - \frac{z}{2}\right)\left(1 - \frac{z}{3}\right)...\left(1 - \frac{z}{n}\right) = \prod_{n=1}^{m}\left(1 - \frac{z}{n}\right) = 0$$

But the equation above only solves for counting numbers; to include 0, multiply by a factor of *z*.

$$z\prod_{n=1}^{m}\left(1 - \frac{z}{n}\right) = 0$$

The above equation only solves for the positive integers, we shall multiply by another product that will also take into account all of the negative integers.

$$z\prod_{n=1}^{m}\left(1-\frac{z}{n}\right)\prod_{n=1}^{m}\left(1+\frac{z}{n}\right)=0$$

(1) $$z\prod_{n=1}^{m}\left(1-\frac{z^2}{n^2}\right)=0$$

The left side of (1) is our function that has all roots contained in the set $\{-m,m\}$ where $m\in\mathbb{Z}$. Now we can establish the following function *G(z)*.

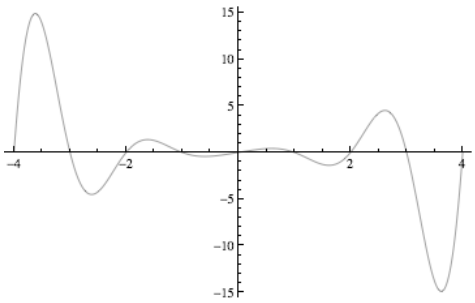
(2) $$G(z)=z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

However, this is not a general expression for a function containing all the integer roots, if we multiply the right hand side of (2) by some function *g(z)*, where *g(z)* has no zeros or singularities, we will obtain every possible function that has all zeros being integers.

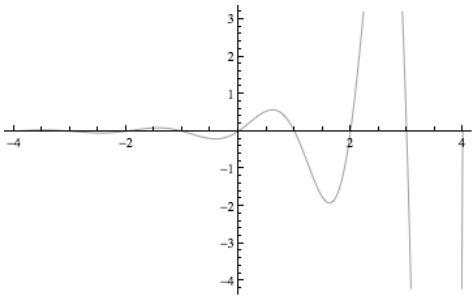
(3) $$G(z)=g(z)z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

Below is a sample table of functions *G*, with different *g* functions and their respective graphs.

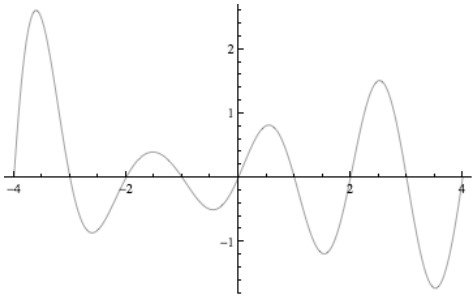
$g(z)=z^2+1$



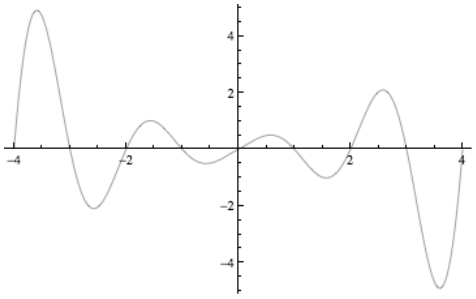
$g(z)=e^z$



$g(z) = \sin(z) + 2$



$g(z) = |z| + 1$



Observe in each graph, that the functions have roots at integer values along the horizontal axis.

Now let us consider the function $f(z) = \sin(\pi z)$, it has the following zeros,

$$z = 0, \pm 1, \pm 2, \pm 3, \pm 4 \ldots \pm n$$

Since this function has all integer zeros, we can use (3) to draw the following conclusion.

$$\sin(\pi z) = g(z) z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

To make sure that the right hand side is normal, we will use exponential functions to express the infinite product of $\sin(\pi z)$,

(4) $$\sin(\pi z) = e^{h(z)} z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

We now need to determine the behavior and value of this unknown function *h(z)*. We shall take the logarithm of (4) and then differentiate to isolate this *h(z).*

$$\log(\sin(\pi z)) = \log\left(e^{h(z)} z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)\right)$$

$$= h(z) + \log(z) + \sum_{n=1}^{\infty} \log\left(1 + \frac{z}{n}\right) + \sum_{n=1}^{\infty} \log\left(1 - \frac{z}{n}\right)$$

Differentiate the above line,

$$\pi\frac{\cos(\pi z)}{\sin(\pi z)}=\pi\cot(\pi z)=h'(z)+\frac{1}{z}+\sum_{n=1}^{\infty}\left(\frac{1/n}{1+z/n}\right)-\sum_{n=1}^{\infty}\left(\frac{1/n}{1-z/n}\right)$$

(5) $$\pi\cot(\pi z)=h'(z)+\frac{1}{z}+\sum_{n=1}^{\infty}\left(\frac{1}{z+n}\right)+\sum_{n=1}^{\infty}\left(\frac{1}{z-n}\right)$$

We can differentiate (5) a second time to obtain,

$$-\frac{\pi^2}{\sin^2(\pi z)}=h''(z)-\frac{1}{z^2}-\sum_{n=1}^{\infty}\left(\frac{1}{(z+n)^2}+\frac{1}{(z-n)^2}\right)$$

The line above can be written in a more compact form that describes $h''(z)$,

(6) $$h''(z)=\sum_{n=-\infty}^{\infty}\left(\frac{1}{(z-n)^2}\right)-\frac{\pi^2}{\sin^2(\pi z)}$$

The question that remains is what is $h''(z)$? Is it constant, 0, or another function on z? Let us consider the two components of (6), the summation and the inverse square sine portion. First, examine the magnitude behavior of the summation,

$$\begin{aligned}\left|\sum_{n=-\infty}^{\infty}\left(\frac{1}{(z-n)^2}\right)\right|&=\left|\sum_{n=-\infty}^{\infty}\left(\frac{1}{z^2-2zn+n^2}\right)\right|\\&=\left|\sum_{n=-\infty}^{\infty}\left(\frac{1}{(x+iy)^2-2(x+iy)n+n^2}\right)\right|\\&=\left|\sum_{n=-\infty}^{\infty}\left(\frac{1}{x^2+i2xy-y^2-2nx-i2yn+n^2}\right)\right|\\&=\left|\sum_{n=-\infty}^{\infty}\left(-\frac{1}{y^2-x^2-n^2+2nx+i(2yn-2xy)}\right)\right|\le 2\sum_{n=0}^{\infty}\left(\frac{1}{y^2+n^2}\right)\end{aligned}$$

From above we can conclude that the magnitude of the summation follows:

(7) $$\left|\sum_{n=-\infty}^{\infty}\left(\frac{1}{(z-n)^2}\right)\right| \le 2\sum_{n=0}^{\infty}\left(\frac{1}{y^2+n^2}\right)$$

Now, we will examine the magnitude of the inverse square sine portion, we will keep in mind that $\sin(z)=\frac{e^{iz}-e^{-iz}}{2i}$.

$$\frac{\pi^2}{\sin^2(\pi z)} = \frac{-4\pi^2}{\left(e^{i\pi z}-e^{-i\pi z}\right)^2}$$
$$= \frac{-4\pi^2}{e^{i2\pi z}+e^{-i2\pi z}-2} = \frac{-4\pi^2}{e^{i2\pi(x+iy)}+e^{-i2\pi(x+iy)}-2}$$
$$= \frac{-4\pi^2}{e^{-2\pi y}e^{i2\pi x}+e^{2\pi y}e^{-i2\pi x}-2} = \frac{-4\pi^2}{e^{-2\pi y}e^{i2\pi x}+e^{2\pi y}e^{-i2\pi x}-2}$$
$$= \frac{-4\pi^2}{e^{-2\pi y}(\cos(2\pi x)+i\sin(2\pi x))+e^{2\pi y}(\cos(2\pi x)-i\sin(2\pi x))-2}$$
$$= \frac{-4\pi^2}{\cos(2\pi x)(e^{-2\pi y}+e^{2\pi y})+i\sin(2\pi x)(e^{-2\pi y}-e^{2\pi y})-2}$$

Now take the magnitude operation on the above line to obtain,

$$\left|\frac{\pi^2}{\sin^2(\pi z)}\right| = \frac{4\pi^2}{\left|\cos(2\pi x)(e^{-2\pi y}+e^{2\pi y})+i\sin(2\pi x)(e^{-2\pi y}-e^{2\pi y})-2\right|}$$
$$= \frac{4\pi^2}{e^{-2\pi y}+e^{2\pi y}-2\cos(2\pi x)} < \frac{4\pi^2}{e^{2\pi|y|}-2}$$

Therefore,

(8) $$\left|\frac{\pi^2}{\sin^2(\pi z)}\right| < \frac{4\pi^2}{e^{2\pi|y|}-2}$$

If we consider the magnitude of $h''(z)$ and lines (8) and (7) we arrive at the following,

$$
\begin{aligned}
|h''(z)| &= \left| \sum_{n=-\infty}^{\infty} \left(\frac{1}{(z-n)^2} \right) - \frac{\pi^2}{\sin^2(\pi z)} \right| \\
&< \left| \sum_{n=-\infty}^{\infty} \left(\frac{1}{(z-n)^2} \right) \right| + \left| \frac{\pi^2}{\sin^2(\pi z)} \right| \\
&< 2\sum_{n=0}^{\infty} \left(\frac{1}{y^2+n^2} \right) + \frac{4\pi^2}{e^{2\pi|y|}-2}
\end{aligned}
$$

Hence,

$$
0 \le |h''(z)| < 2\sum_{n=0}^{\infty} \left(\frac{1}{y^2+n^2} \right) + \frac{4\pi^2}{e^{2\pi|y|}-2}
$$

If we allow *y* on the right hand side to approach infinite, we can conclude the following.

$$
\begin{aligned}
0 \le |h''(z)| &< \lim_{y\to\infty} \left\{ 2\sum_{n=0}^{\infty} \left(\frac{1}{y^2+n^2} \right) + \frac{4\pi^2}{e^{2\pi|y|}-2} \right\} \\
0 \le |h''(z)| &< 0+0 \\
0 \le |h''(z)| &< 0
\end{aligned}
$$

The above statement implies that the magnitude of $h''(z)$ is 0, and if the magnitude is 0, then the actual function $h''(z) = 0$. With this mind we can conclude that $h'(z) = c$, where *c* is a constant and substitute this into (5), also combine the two summations in (5).

(9) $$\pi \cot(\pi z) = c + \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{2z}{z^2-n^2} \right)$$

If we let $z = -z$ in (9) we obtain the following,

(10) $$-\pi \cot(\pi z) = c - \frac{1}{z} - \sum_{n=1}^{\infty} \left(\frac{2z}{z^2-n^2} \right)$$

Combining lines (9) and (10) by adding them together gives $2c = 0$, therefore $h'(z) = 0$. This means that $h(z) = \text{constant}$, this also implies that $e^{h(z)} = \text{constant} = k$. If we substitute the constant into (4), we get the following,

$$\sin(\pi z) = kz\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

$$\frac{\sin(\pi z)}{z} = k\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

Take the limit as *z* approaches 0 yields the following,

$$\pi\lim_{z\to 0}\left\{\frac{\sin(\pi z)}{\pi z}\right\} = k\lim_{z\to 0}\left\{\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)\right\}$$

$$\pi = k\cdot(1-0)\cdot(1-0)\cdot\underbrace{(1-0)...(1-0)}_{\text{infinite product of 1's}}\cdot(1-0)$$

Therefore,

(11) $$\sin(\pi z) = \pi z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

PART 2: $\cos(\pi z)$

Let us now consider, $\cos(\pi z)$, it has zeros at $z = \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, ..., \pm\frac{2n+1}{2}$, where $n \in \{0,1,2,3...\}$. Therefore, utilizing the same techniques as in Part 1 we obtain the following,

(12) $$\cos(\pi z) = e^{h(z)}\prod_{n=0}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right)$$

Next we will take the logarithm of both sides of (12).

$$\log(\cos(\pi z)) = h(z) + \sum_{n=0}^{\infty}\left(\log\left(1-\frac{2z}{2n+1}\right)+\log\left(1+\frac{2z}{2n+1}\right)\right)$$

If differentiate the line above we obtain,

$$\begin{aligned}-\pi\tan(\pi z) &= h'(z) + \sum_{n=0}^{\infty}\left(\frac{-1/(2n+1)}{1-2z/(2n+1)}+\frac{1/(2n+1)}{1+2z/(2n+1)}\right)\\ &= h'(z) + \sum_{n=0}^{\infty}\left(\frac{1}{2n+1+2z}-\frac{1}{2n+1-2z}\right)\\ &= h'(z) + \sum_{n=0}^{\infty}\left(\frac{1}{2z+(2n+1)}+\frac{1}{2z-(2n+1)}\right)\end{aligned}$$

Therefore,

(13) $$-\pi\tan(\pi z) = h'(z) + \sum_{n=0}^{\infty}\left(\frac{1}{2z+(2n+1)}+\frac{1}{2z-(2n+1)}\right)$$

Differentiate a second time to obtain,

$$\begin{aligned}\frac{-\pi^2}{\cos^2(\pi z)} &= h''(z) - 2\sum_{n=0}^{\infty}\left(\frac{1}{\left(2z+(2n+1)\right)^2}+\frac{1}{\left(2z-(2n+1)\right)^2}\right)\\ &= h''(z) - 2\sum_{n=-\infty}^{\infty}\frac{1}{\left(2z-(2n+1)\right)^2}\end{aligned}$$

From the above line we can conclude,

(14) $$h''(z) = 2\sum_{n=-\infty}^{\infty}\frac{1}{\left(2z-(2n+1)\right)^2} - \frac{\pi^2}{\cos^2(\pi z)}$$

Let us examine the magnitude behavior of the summation portion using comparison tests.

$$\left|2\sum_{n=-\infty}^{\infty}\frac{1}{\left(2z-(2n+1)\right)^2}\right| = 2\left|\sum_{n=-\infty}^{\infty}\frac{1}{\left(2z-(2n+1)\right)^2}\right|$$
$$= 2\left|\sum_{m=odd}^{\{-\infty,\infty\}}\frac{1}{\left(2z-m\right)^2}\right|$$
$$< 2\sum_{m=odd}^{\{-\infty,\infty\}}\frac{1}{4y^2+m^2}$$

From the above lines we can conclude,

(15) $$\left|2\sum_{n=-\infty}^{\infty}\frac{1}{\left(2z-(2n+1)\right)^2}\right| < 2\sum_{m=odd}^{\{-\infty,\infty\}}\frac{1}{4y^2+m^2} = 4\sum_{n=0}^{\infty}\frac{1}{4y^2+(2\mathrm{n}+1)^2}$$

Let us now examine the cosine term from (14).

$$\frac{-\pi^2}{\cos^2(\pi z)} = \frac{-4\pi^2}{\left(e^{i\pi z}+e^{-i\pi z}\right)^2} = \frac{-4\pi^2}{\left(e^{i\pi(x+iy)}+e^{-i\pi(x+iy)}\right)^2}$$
$$= \frac{-4\pi^2}{\left(e^{i\pi x-\pi y}+e^{-i\pi x+\pi y}\right)^2}$$
$$= \frac{-4\pi^2}{\left(e^{-\pi y}\left(\cos(\pi x)+i\sin(\pi x)\right)+e^{\pi y}\left(\cos(\pi x)-i\sin(\pi x)\right)\right)^2}$$
$$= \frac{-4\pi^2}{\left(\cos(\pi x)\left(e^{\pi y}+e^{-\pi y}\right)-i\sin(\pi x)\left(e^{\pi y}-e^{-\pi y}\right)\right)^2}$$

If we take the magnitude of the above lines we obtain,

$$\left|\frac{-\pi^2}{\cos^2(\pi z)}\right| = \frac{4\pi^2}{e^{2\pi y}+e^{-2\pi y}+2\cos(2\pi x)} < \frac{4\pi^2}{e^{2\pi|y|}+2}$$

Hence,

(16) $$\left|\frac{-\pi^2}{\cos^2(\pi z)}\right| < \frac{4\pi^2}{e^{2\pi|y|}+2}$$

Combining (14)-(16) we can conclude the following,

$$\begin{aligned}
|h''(z)| &= \left| 2\sum_{n=-\infty}^{\infty} \frac{1}{\left(2z-(2n+1)\right)^2} - \frac{\pi^2}{\cos^2(\pi z)} \right| \\
&< \left| 2\sum_{n=-\infty}^{\infty} \frac{1}{\left(2z-(2n+1)\right)^2} \right| + \left| \frac{\pi^2}{\cos^2(\pi z)} \right| \\
&< 4\sum_{n=0}^{\infty} \frac{1}{4y^2+(2n+1)^2} + \frac{4\pi^2}{e^{2\pi|y|}+2} \quad \therefore \\
|h''(z)| &< 4\sum_{n=0}^{\infty} \frac{1}{4y^2+(2n+1)^2} + \frac{4\pi^2}{e^{2\pi|y|}+2}
\end{aligned}$$

For the line above, if we let *y* become arbitrarily large we get,

$$\begin{aligned}
0 \le |h''(z)| &< \lim_{y\to\infty} \left\{ 4\sum_{n=0}^{\infty} \frac{1}{4y^2+(2n+1)^2} + \frac{4\pi^2}{e^{2\pi|y|}+2} \right\} \\
0 \le |h''(z)| &< 0+0 \\
0 \le |h''(z)| &< 0
\end{aligned}$$

This means that $h''(z)$ converges absolutely to 0, and we can conclude that $h'(z) = \text{constant}$.

Let us now solve for $h'(z)$, if we recall from (13),

(17) $$-\pi\tan(\pi z) = C + \sum_{n=0}^{\infty}\left(\frac{4z}{4z^2-(2n+1)^2} \right)$$

If we let z = -z,

(18) $$\pi\tan(\pi z) = C - \sum_{n=0}^{\infty}\left(\frac{4z}{4z^2-(2n+1)^2} \right)$$

If we combine (17) and (18) by addition, we get *2C = 0*. Therefore, $h'(z) = 0$ and $h(z) = \text{constant} = k$. If we substitute this back into (12) and take the limit as z approaches 0 we obtain the following,

$$\lim_{z\to 0}\{\cos(\pi z)\} = k\lim_{z\to 0}\left\{\prod_{n=0}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right)\right\}$$

$$1 = k\cdot(1-0)\cdot\underbrace{(1-0)\cdots(1-0)}_{\text{infinite product of 1's}}\cdot(1-0)$$

Therefore, $k = 1$.

$$\cos(\pi z) = \prod_{n=0}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right)$$

PART 3: $\tan(\pi z)$ and $\cot(\pi z)$

Let us now consider the tangent function, in order to derive its infinite product expansion we will use the expansions from Parts 1 and 2.

(19) $$\sin(\pi z) = \pi z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)$$

(20) $$\cos(\pi z) = \prod_{n=0}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right) = \left(1-4z^2\right)\prod_{n=1}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right)$$

We will take lines (19) and (20) and divide them to get a product expansion for the tangent function.

$$\tan(\pi z) = \frac{\sin(\pi z)}{\cos(\pi z)}$$

$$= \frac{\pi z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)}{\left(1-4z^2\right)\prod_{n=1}^{\infty}\left(1-\frac{4z^2}{(2n+1)^2}\right)}$$

$$= \frac{\pi z}{\left(1-4z^2\right)}\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right)\left(1-\frac{4z^2}{(2n+1)^2}\right)^{-1}$$

Therefore,

(21) $$\tan(\pi z) = \frac{\pi z}{\left(1-4z^2\right)} \prod_{n=1}^{\infty} \left(1-\frac{z^2}{n^2}\right)\left(1-\frac{4z^2}{(2n+1)^2}\right)^{-1}$$

To find the cotangent function, we shall take the reciprocal of (21).

(22) $$\cot(\pi z) = \frac{\left(1-4z^2\right)}{\pi z} \prod_{n=1}^{\infty} \left(1-\frac{z^2}{n^2}\right)^{-1}\left(1-\frac{4z^2}{(2n+1)^2}\right)$$

Part 4: Applying Special Products To Find Square Roots

Now that we have established our infinite product expansions for four trigonometric functions in the complex plane, let us utilize them to find expansions for square roots. We know that $\sin(\pi/2) = 1$; this means that $z = ½$.

$$\sin(\pi/2) = \frac{\pi}{2} \prod_{n=1}^{\infty} \left(1-\frac{1}{4n^2}\right) = 1$$

From the above line we can conclude that,

$$\begin{aligned}
\frac{\pi}{2} &= \prod_{n=1}^{\infty} \left(1-\frac{1}{4n^2}\right)^{-1} \\
&= \prod_{n=1}^{\infty} \left(\frac{4n^2-1}{4n^2}\right)^{-1} \\
&= \prod_{n=1}^{\infty} \left(\frac{4n^2}{4n^2-1}\right) = \prod_{n=1}^{\infty} \left(\frac{2n}{2n-1} \cdot \frac{2n}{2n+1}\right)
\end{aligned}$$

This equals the Wallis Product.

Let us now consider $z = ¼$, with this substitution $\sin(\pi/4) = \frac{\sqrt{2}}{2}$.

(23) $$\sin(\pi/4) = \frac{\pi}{4}\prod_{n=1}^{\infty}\left(1-\frac{1}{16n^2}\right) = \frac{\sqrt{2}}{2}$$

If we multiply both sides of (23) by 2 and combine this with the Wallis Product we get,

$$\begin{aligned}\sqrt{2} &= \frac{\pi}{2}\prod_{n=1}^{\infty}\left(1-\frac{1}{16n^2}\right)\\ &= \left\{\prod_{n=1}^{\infty}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right)\right\}\cdot\left\{\prod_{n=1}^{\infty}\left(1-\frac{1}{16n^2}\right)\right\}\\ &= \left\{\prod_{n=1}^{\infty}\left(\frac{4n^2}{4n^2-1}\right)\right\}\cdot\left\{\prod_{n=1}^{\infty}\left(\frac{16n^2-1}{16n^2}\right)\right\}\\ &= \prod_{n=1}^{\infty}\left(\frac{16n^2-1}{4\left(4n^2-1\right)}\right)\end{aligned}$$

Hence,

(24) $$\sqrt{2} = \prod_{n=1}^{\infty}\left(\frac{16n^2-1}{4\left(4n^2-1\right)}\right) = \prod_{n=1}^{\infty}\left(1+\frac{3}{4(4n^2-1)}\right)$$

Let us now consider $\sin\left(\pi/3\right)$,

(25) $$\sin(\pi/3) = \frac{\pi}{3}\prod_{n=1}^{\infty}\left(1-\frac{1}{9n^2}\right) = \frac{\sqrt{3}}{2}$$

If we multiply (25) by two 2 on both sides of the equality we obtain the following.

$$\sqrt{3} = \frac{2}{3}\pi\prod_{n=1}^{\infty}\left(1-\frac{1}{9n^2}\right)$$

Now we replace π with the Wallis Product we get,

$$\sqrt{3} = \frac{4}{3}\prod_{n=1}^{\infty}\left(\frac{4n^2}{4n^2-1}\right)\prod_{n=1}^{\infty}\left(1-\frac{1}{9n^2}\right)$$
$$= \frac{4}{3}\prod_{n=1}^{\infty}\left(\frac{4n^2}{4n^2-1}\cdot\frac{9n^2-1}{9n^2}\right)$$
$$= \frac{4}{3}\prod_{n=1}^{\infty}\left(\frac{36n^2-4}{36n^2-9}\right)$$
$$= \frac{4}{3}\prod_{n=1}^{\infty}\left(1+\frac{5}{9(4n^2-1)}\right)$$

Therefore,

(26) $$\sqrt{3} = \frac{4}{3}\prod_{n=1}^{\infty}\left(1+\frac{5}{9(4n^2-1)}\right)$$

If we multiply (24) and (26) we get,

$$\sqrt{2}\sqrt{3} = \left\{\prod_{n=1}^{\infty}\left(1+\frac{3}{4(4n^2-1)}\right)\right\}\left\{\frac{4}{3}\prod_{n=1}^{\infty}\left(1+\frac{5}{9(4n^2-1)}\right)\right\}$$
$$= \frac{4}{3}\prod_{n=1}^{\infty}\left\{\left(1+\frac{3}{4(4n^2-1)}\right)\left(1+\frac{5}{9(4n^2-1)}\right)\right\}$$
$$= \frac{4}{3}\prod_{n=1}^{\infty}\left(\frac{\left(9n^2-1\right)\left(16n^2-1\right)}{9\left(4n^2-1\right)^2}\right)$$

Therefore,

(27) $$\sqrt{6} = \frac{4}{3}\prod_{n=1}^{\infty}\left(\frac{\left(9n^2-1\right)\left(16n^2-1\right)}{9\left(4n^2-1\right)^2}\right)$$

These product series expansions of the square roots provide us with a more efficient way of calculating them, instead of using summation series expansions.

REFERENCES

[1] Knopp, K., Bagemihl, F. (translator) (**1996**) *Theory of Functions: Parts 1 and 2*, Dover Publications Inc. (New York).